

# Design and evaluation of a robust and explainable intrusion detection framework for 5G/B5G networks

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## ABSTRACT

As Fifth-generation (5G) networks advance towards future 6G technology, they provide enhanced capabilities in terms of data rates, latency, and connectivity. The expanding and heterogeneous landscape further extends the attack surface with increasing vulnerability to a wide range of dynamic cyber threats. Traditional Intrusion Detection Systems, relying on static signature-based methods, face significant challenges in detecting novel attacks in such complex environments. This paper proposes an explainable hybrid AE-XGBoost framework to address these gaps. In this proposed framework, the deep autoencoder serves as the backbone for two primary tasks: learning robust features from high-dimensional traffic data and synthesizing high-quality samples for minority, underrepresented attack classes. It effectively forms a class-balanced dataset. This is further extended to integrate Shapley Additive Explanations (SHAP), which provide needed interpretability. The framework robustness is determined by an extensive evaluation strategy. Extensive experiments are conducted using a 5-fold cross-validation and cross-dataset validation using three different datasets. The experimental results demonstrate the model effectiveness and promising generalization across the different datasets. Across these settings, the proposed IDS achieves F1-scores between approximately 0.96 and 0.999 and AUC values above 0.97, with only minor performance differences between original and synthetic test samples. SHAP-based analysis further reveals that a small subset of 5G- and TCP-level features consistently dominates the predictions, offering actionable explanations for security analysts. This provides a successful, scalable, and reliable security measure that proves effective and appropriate for 6G mobile network implementations in the near future.

## 1. Introduction

Fifth-generation (5G) networks are the edge of contemporary telephony, featuring high data speeds, ultra-low latency, and a much higher density of connectivity relative to past cellular generations. 5G networks also raise the attack surface and pose advanced security problems. Classical Intrusion Detection Systems (IDS), which make use of pre-known signatures or heuristic-based schemes, might encounter difficulties in recognizing zero-day attacks, stealthy exploits, or advanced polymorphic payloads. Recent studies proposed the use of Machine Learning (ML) and Deep Learning (DL) techniques to combat the dynamic character of 5G intrusion detection [1,2]. These methods have shown significant potential in learning advanced features end-to-end from raw network traffic.

5G environments, however, pose additional challenges that infringe detection success. Firstly, the 5G traffic high-dimensionality with typically millions of flows and enormous feature spaces, i.e., packet-level, flow-level, and session-level information, can overwhelm conventional machine learning methods [3–5]. Secondly, class imbalance remains a significant challenge in intrusion detection systems, particularly because certain sophisticated attack types, such as slow-rate Denial-of-Service (DoS) attacks or targeted scanning, occur far less frequently compared to high-volume, brute-force attacks. This disparity in frequency leads to heavily skewed datasets, where machine learning models tend to overfit to majority classes, resulting in poor detection performance for rare but potentially more damaging attack patterns [6–9].

In order to solve these problems, this paper proposes an integrated intrusion detection system with two primary components. The first one

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is a deep autoencoder (AE) [10], which serves multiple purposes: it maps high-dimensional data to low-dimensional latent representations, employs reconstruction based techniques to handle anomalies, and produces additional minority-class samples to address class imbalance. The second attribute is XGBoost, an effective gradient boosting algorithm that has been proven to exhibit high accuracy in classification tasks on tabular data. Through the integration of deep autoencoder-induced representation learning and the combination of XGBoost [11], the system adequately manages the high dimensional and class-imbalanced nature of 5G traffic. The procedure includes data pre-processing, dimensionality reduction, and synthetic generation of underrepresented attack class samples by training the autoencoder, followed by classification using XGBoost. The model is tested with three datasets: 5G-NIDD [12] and CIC-IDS2017 [13,14], and the 5G dataset [15]. Cross-data set experiments are performed to demonstrate the generalizability of the model when exposed to traffic from multiple sources. The main contributions of this paper are summarized as follows:

- A novel hybrid Autoencoder-XGBoost framework is proposed, which uniquely leverages the autoencoder for both deep feature extraction and synthetic sample generation, effectively addressing the dual challenges of high dimensionality and severe class imbalance in network traffic data.
- Shapley Additive Explanations (SHAP) are integrated to improve model transparency, offering essential local and global interpretability for model predictions, a vital aspect of trustworthy security systems.
- A thorough evaluation of the framework is performed using three distinct datasets (5G-NIDD, CIC-IDS2017, and 5G Dataset), including cross-dataset validation to demonstrate the model's strong generalizability and effectiveness in managing heterogeneous traffic from multiple, unseen sources.

The remainder of the paper is structured as follows. Section 2 reviews related work in 5G intrusion detection, autoencoder-based security methods, and synthetic data augmentation. While Section 3 describes the proposed hybrid pipeline, highlighting differences from the baseline approach, Section 4 details the implementation. Section 5 presents empirical results, including a comparison of detection results before and after using the enhanced method. Section 6 summarizes the work and suggests directions for further research.

## 2. Background and related work

The complexity of 5G security and the growing sophistication of network attacks required researchers to embark on extensive investigations, embedding ML and DL techniques into efforts to detect intrusions more objectively. Modern 5G-enabled networks are characterized by their ability to support a wide range of service types, including eMBB, mMTC, and URLLC [1,4]. These service types introduce diverse traffic patterns with different latency, throughput, and reliability requirements, making it even more challenging to detect anomalous or malicious behavior in real-time. As these networks began to grow and develop, it became even more challenging for traditional signature-based IDS to evolve and adequately capture them, in large part due to these signatures being pre-defined attack patterns that lacked sufficient detection support for new threats. This had been accompanied by reliance on more intelligent, data-aware models capable of detecting threats by learning and generalizing from clutter patterns [16]. To tackle these challenges, researchers have increasingly focused on examining the use of ML-based strategies, such as Random Forests, Support Vector Machines (SVMs), and other neural networks [17–20]. These methods offer the promise of detecting intricate and non-linear relationships in network traffic and learning to adapt to evolving threat profiles.

Despite all their potential, however, ML and DL models are not without their flaws. One of the most significant vulnerabilities is sus-

ceptibility to advanced and covert attacks that are tailored to merge with high traffic levels. These insidious intrusions can evade detection mechanisms that are not temporal or contextually aware of the network environment. Additionally, the record-breaking high dimensionality and dynamics of 5G traffic also exacerbate the challenge in model training, necessitating even more scalable, interpretable, and cost-efficient solutions that can be accommodated within highly strict latency constraints. Therefore, recent research efforts to enhance the detection accuracy with concurrent exploration of more explainable and adaptive solutions for IDS under the 5G environment are in higher [21,22].

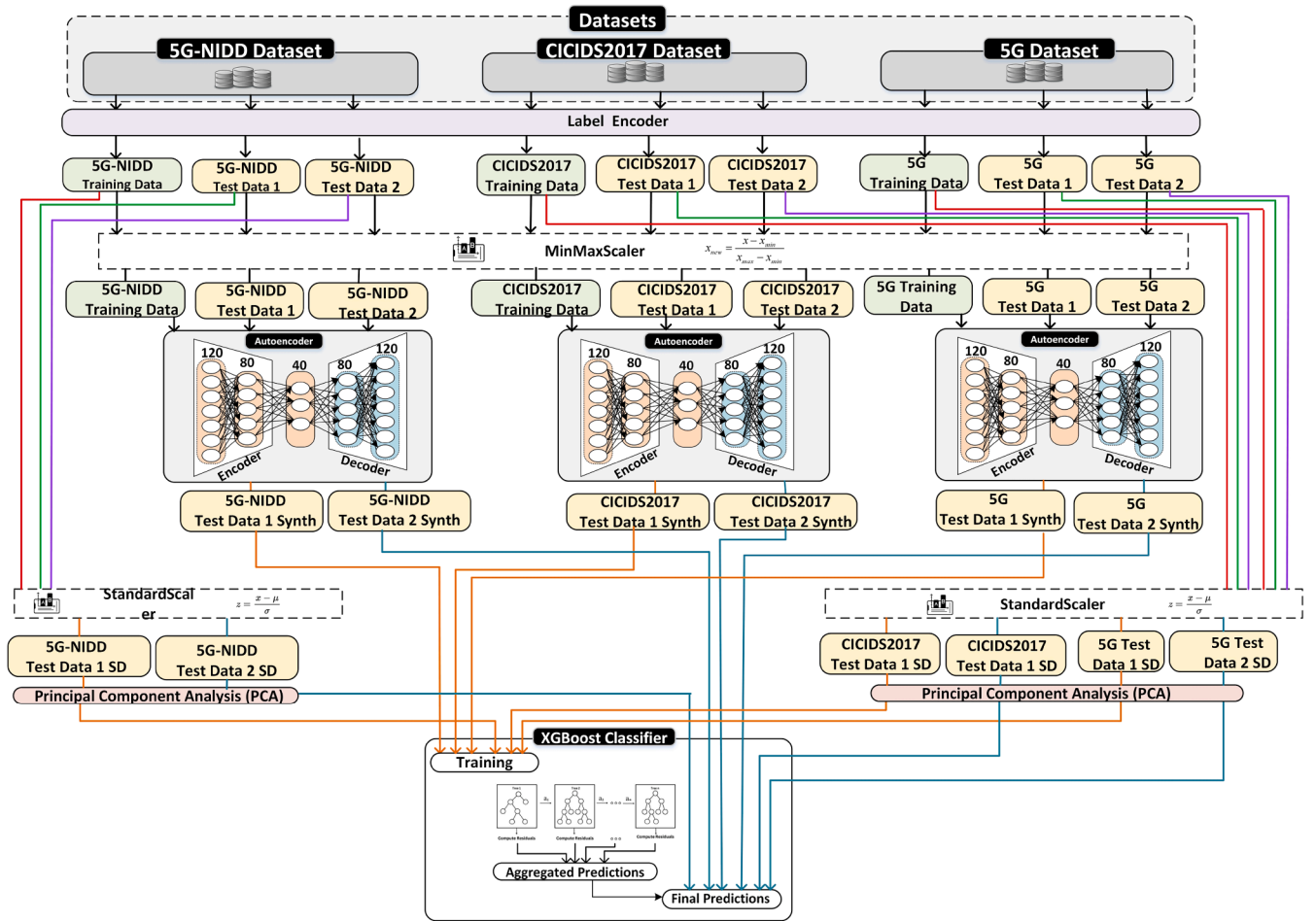
Recent work has highlighted the effectiveness of autoencoders for network security. An autoencoder maps the data into a lower-dimensional latent space by an encoder and then maps it back by a decoder, and the reconstruction loss can be used as a natural anomaly signal. An IDS utilizing an autoencoder and XGBoost for feature extraction and also classification was proposed in [23], and the results on the CSE-CIC-IDS2018 dataset were improved [24]. used Variational Autoencoders (VAE) and XGBoost with class-wise focal loss for handling extreme class imbalance, while other works put together stacked sparse autoencoders and XGBoost for optimizing the usage of learned representations [25]. Besides traditional autoencoders, there are also extensions in the form of conditional, variational, and even cellular autoencoders, which might be harder to train but have rich features for generating domain-specific synthetic samples. XGBoost has been a successful gradient-boosting platform that is well-suited to tabular, high-dimensional data and thus naturally compatible with IDS applications. Its recursive tree construction and sophisticated regularization techniques give it stability and scalability, and give it a way to fight the noise and imbalance of network traffic in real life effectively [11]. Class imbalance remains a very bad problem in the area of cybersecurity, particularly when stealth or zero-day attacks are grossly underrepresented during training. Classic oversampling approaches, such as SMOTE, alleviate this to some extent, but at times do not replicate more subtle behavior in high-dimensional feature spaces.

Autoencoders can recover minority class samples and perturb them slightly to create more realistic synthetic data and thus augment the training set for better classification performance. With the increasing size and complexity of network infrastructure, especially in the context of next-generation 6G networks, the need for intelligent and transparent intrusion detection systems has become all the more imperative. Conventional machine learning IDS models, even though highly performant, are black boxes, and hence, security analysts find it challenging to understand their decision-making process. This transparency results in serious barriers to trust, flexibility, and deployment in operations. To address this, the adoption of explainable artificial intelligence (XAI) in IDS architectures has received considerable momentum, seeking to fill the gap between interpretability and high performance [26,27].

In [28], Random Forest (RF) [29], Decision Trees [30], and Support Vector Machines (SVM) [31] ensemble classifiers have been used with a voting mechanism to enhance accuracy and minimize false positives and attained around 96.25% accuracy on the CIC-IDS 2017 dataset. Explainability in this case is dealt with using methods such as Local Interpretable Model-agnostic Explanations (LIME) [32], which gives transparency to the model's decision-making process. In attempts to make local and global interpretability better, certain XAI methods like SHAP, Permutation Importance (PI) [33], LIME, and contextual importance and utility scores have been integrated into tree-based machine learning models like XGBoost, RF, and Light Gradient Boosting Machine (LGBM) [34]. In [35], feature selection with PI as the basis gave a subset of 15 best features, and LGBM attained 92% accuracy on the NSL-KDD dataset [36]. To provide a structured overview, Table 1 summarizes related works, highlighting the specific techniques employed and the limitations that remain unaddressed. As observed, while many studies utilize Deep Learning for feature extraction, few frameworks simultaneously address the high-dimensionality of 5G traffic, the

**Table 1**  
Comparison of the Proposed Framework with Existing Approaches.

| Category                  | Technique                          | Key Strengths                                                                                                                | Limitations Addressed                                                                                                                                         |
|---------------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Traditional ML [17–20]    | SVM, Random Forest, Decision Trees | Low computational complexity.                                                                                                | Manual feature engineering required; poor detection of zero-day attacks.                                                                                      |
| AE + XGBoost [21–23]      | Autoencoder features + XGBoost     | Robust feature compression; improved accuracy.                                                                               | Often lacks mechanisms for synthetic minority oversampling; Without XAI integration.                                                                          |
| VAE [24]                  | Variational AEs                    | Generates synthetic data to handle class imbalance.                                                                          | High training instability; high computational cost, unsuitable for low-latency 5G edge deployment.                                                            |
| Ensemble + XAI [26–28,35] | Voting Classifiers + LIME          | High accuracy on tabular data; provides local interpretability.                                                              | Does not leverage deep representation learning for complex traffic patterns; Computationally expensive for real-time streams.                                 |
| <b>Proposed Framework</b> | <b>Hybrid AE + XGBoost + SHAP</b>  | Deep Feature Learning, latent-space class-imbalance mitigation, and SHAP-based global/local explanations on 5G/B5G datasets. | <b>Solution:</b> Efficiently handles high-dimensional 5G data, corrects severe class imbalance with low-cost synthesis, and ensures trust via explainability. |



**Fig. 1.** Overview of the proposed Hybrid Autoencoder-XGBoost (AE-XGB) intrusion detection pipeline for 5G networks. Raw network traffic is first preprocessed and normalized, then passed through a deep AE that (i) learns compact latent representations of high-dimensional flows and (ii) generates synthetic samples for minority attack classes in latent space. The balanced latent features are then fed to an XGBoost classifier to produce the final anomaly predictions.

need for high-fidelity synthetic data generation, and analyst-friendly explainability.

Despite these developments, challenges in the deployment of autoencoder-based IDS on large 5G networks exist. Highly deep model training utilizing massive datasets is computationally costly, while extreme class imbalance and interpretability requirements make real-time deployment challenging. This paper evades these shortcomings in the sense that (1) it uses a well-engineered deep autoencoder for compact feature representation and synthetic sample synthesis, (2) it uses XGBoost for stable classification, and (3) it performs cross-dataset experiments to ensure the proposed pipeline's generality.

### 3. Methodology

To effectively address the challenges of high dimensionality and significant class imbalance in 5G intrusion detection, this paper introduces a multi-stage, hybrid framework. The primary objective is to establish a robust pipeline that is both accurate and capable of handling uneven data distributions. It combines a deep Autoencoder (AE) with an XGBoost classifier to create a reliable and precise pipeline. The AE converts high-dimensional network traffic into a lower-dimensional representation and generates synthetic samples for underrepresented attack classes.

Fig. 1 depicts the architecture of the proposed pipeline, referred to as the Hybrid Autoencoder-XGBoost IDS Framework. In this section, the pipeline is thoroughly described, with each component and its functions detailed extensively. This is then followed by a mathematical formulation of the framework, establishing a solid theoretical foundation that aids in the understanding, analysis, and implementation of the proposed method.

### 3.1. Step 1: Data preprocessing and initial encoding

Raw network traffic datasets from CSV files often contain noisy, redundant, and inconsistent data. Therefore, a thorough preprocessing stage is crucial before model training to ensure data quality and enhance performance. This process typically involves managing missing values by either removing erroneous entries or imputing them using statistical methods. Additionally, feature scaling normalizes numerical features to prevent larger magnitudes from skewing results. Finally, categorical features must be encoded to convert non-numeric data, such as protocol types and service names, into a machine-readable format.

Let  $\mathbf{X} \in \mathbb{R}^{N \times D}$  denote the original dataset, where  $N$  represents the total number of traffic samples (rows) (i.e., the total number of network flows/packets) and  $D$  corresponds to the number of features per sample (columns). MinMax scaling or normalization by simple range is applied to get the transformed dataset  $\tilde{\mathbf{X}}$ . Categorical features (e.g., protocols) are one-hot or label-encoded.

### 3.2. Step 2: Autoencoder feature compression and reconstruction

A deep autoencoder is employed and adapted to encode each normalized sample  $\tilde{\mathbf{x}}_i$  into a latent code  $\mathbf{z}_i$  of dimension  $k \ll D$ :

$$\mathbf{z}_i = e_\theta(\tilde{\mathbf{x}}_i), \quad i = 1, \dots, N, \quad (1)$$

where  $e_\theta(\cdot)$  is the encoder parameterized by  $\theta$ . The decoder  $d_\phi(\cdot)$  reconstructs  $\tilde{\mathbf{x}}_i$ :

$$\hat{\mathbf{x}}_i = d_\phi(\mathbf{z}_i) = d_\phi(e_\theta(\tilde{\mathbf{x}}_i)). \quad (2)$$

The autoencoder's internal objective (reconstruction error) is:

$$\mathcal{L}_{\text{AE}}(\theta, \phi) = \sum_{i=1}^N \|\tilde{\mathbf{x}}_i - \hat{\mathbf{x}}_i\|^2. \quad (3)$$

**Shared Representation for Classification.** Critically, learned latent representations  $\{\mathbf{z}_i\}$  are directly passed into a classification sub-objective. That is, having once learned,  $\mathbf{z}_i$  is the primary feature vector for XGBoost.  $\theta$  is therefore not merely minimizing (3) in isolation but does so at the same time as it conforms to the requirements of the classifier.

### 3.3. Step 3: Synthetic data generation

After training the AE, class imbalance is addressed by generating synthetic samples for minority-class attacks. Let  $\mathbf{x}_m$  represent a minority-class sample with latent code  $\mathbf{z}_m = e_\theta(\mathbf{x}_m)$ . A new synthetic sample is produced as follows:

$$\tilde{\mathbf{x}}_m = d_\phi(\mathbf{z}_m + \epsilon), \quad (4)$$

where  $\epsilon \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$  is small Gaussian noise. Only those  $\tilde{\mathbf{x}}_m$  with acceptably low reconstruction error or domain-consistent features are kept. This procedure amplifies the minority attack class, making the final training distribution more balanced. It is important to note that the objective of this synthetic data generation strategy is not to create new or complex attack behaviors, but to introduce controlled variability around existing samples. Gaussian perturbations in the latent space capture natural traffic variability while preserving core attack characteristics. Attacks are modeled through learned feature representations rather than packet-level construction, and the generated samples act as a data augmentation mechanism to mitigate class imbalance and improve robustness, rather than replacing real attack traffic.

While more advanced approaches such as Conditional VAEs and GANs can generate richer and more diverse samples, they typically introduce higher computational cost, training instability, and additional design complexity. In contrast, the adopted latent-space perturbation provides a lightweight, stable, and efficient augmentation strategy aligned with the practical objectives of this work.

### 3.4. Step 4: XGBoost classification with embedded AE representations

Let  $\tilde{\mathbf{X}} \in \mathbb{R}^{(N+M) \times k}$  be the final training features set, where  $M$  is the number of synthetic samples generated. Each row  $\tilde{\mathbf{x}}_i$  represents the latent code  $\mathbf{z}_i$  (or  $\tilde{\mathbf{z}}_i$  for synthetic data).

XGBoost builds a sequence of regression trees  $\{f_t\}_{t=1}^T$ , summing their predictions to give the predicted log-odds for sample  $i$ :

$$\hat{y}_i = \sum_{t=1}^T f_t(\tilde{\mathbf{x}}_i). \quad (5)$$

A logistic loss is employed for binary or multi-class intrusion detection. The XGBoost objective can be written as:

$$\mathcal{L}_{\text{XGB}} = \sum_{i=1}^{N+M} \ell(\hat{y}_i, y_i) + \sum_{t=1}^T \Omega(f_t), \quad (6)$$

where  $\ell$  is the logistic loss and  $\Omega$  imposes regularization on the trees.

## 4. Implementation and experimental setup

### 4.1. Datasets

To demonstrate the robustness, efficiency, and generalizability of the proposed methodology, three distinct and contemporary datasets were employed: 5G-NIDD, CIC-IDS2017, and the 5G dataset. These datasets provide comprehensive packet capture files (.pcap) and corresponding labels; however, training and evaluation are conducted using features extracted from the PCAP files in CSV format.

#### 4.1.1. 5G-NIDD Dataset

The 5G-NIDD Dataset is a new dataset designed to detect intrusions in 5G networks. It was created using the 5G Test Network (5GTN) at the University of Oulu in Finland. It captures traffic from real 5G base stations (gNB) and user equipment (UE), providing a realistic representation of 5G control and data plane traffic under various attack scenarios. Benign traffic was generated by real mobile devices attached to the 5GTN, producing HTTP/HTTPS web browsing and live streaming, as well as SSH and SFTP sessions. Eight different attack types were launched against this infrastructure (ICMP, UDP, and SYN floods, HTTP flood, slow-rate DoS, SYN scan, TCP connect scan, and UDP scan). Malicious and Benign traffic were recorded from gateway nodes in two base stations [12]. The dataset is comprehensive, containing **1,215,890 samples**, each with **48 features**. The complete feature list is provided in Table 2.

#### 4.1.2. CIC-IDS2017 Dataset

The CIC-IDS2017 dataset was developed by the Canadian Institute in a controlled enterprise-like network for Cybersecurity and has become a benchmark dataset in the domain of network intrusion detection. The environment consists of an internal victim network with 12 machines running various operating systems (Windows, Ubuntu, Mac OS X), and an external attacker network including Kali Linux and Windows hosts. Realistic benign traffic was generated for 25 user profiles using the B-Profile system over HTTP, HTTPS, FTP, SSH, and Email. Traffic was captured from a mirror port on the main switch over five days (July 3-7, 2017), with Monday consisting only of benign traffic, and the remaining days featuring a mix of benign activity and various attacks, including brute-force, DoS, DDoS, web attacks, infiltration, botnet, and port scans [13]. Each attack type is saved in an individual .pcap files, and labeling



**Table 2**

5G-NIDD Attack Dataset for Comparison of features across three datasets, features in blue represent common attributes.

| No. | Feature Name | No. | Feature Name | No. | Feature Name | No. | Feature Name              |
|-----|--------------|-----|--------------|-----|--------------|-----|---------------------------|
| 1   | Dur          | 13  | dTtl         | 25  | dMeanPktSz   | 37  | DstRate                   |
| 2   | RunTime      | 14  | sHops        | 26  | Load         | 38  | State                     |
| 3   | Mean         | 15  | dHops        | 27  | SrcLoad      | 39  | SrcWin                    |
| 4   | Sum          | 16  | Cause        | 28  | DstLoad      | 40  | DstWin                    |
| 5   | Min          | 17  | TotPkts      | 29  | Loss         | 41  | sVid                      |
| 6   | Max          | 18  | SrcPkts      | 30  | SrcLoss      | 42  | dVid                      |
| 7   | Proto        | 19  | DstPkts      | 31  | DstLoss      | 43  | SrcTCPBase                |
| 8   | sTos         | 20  | TotBytes     | 32  | pLoss        | 44  | DstTCPBase                |
| 9   | dTos         | 21  | SrcBytes     | 33  | SrcGap       | 45  | TcpRtt                    |
| 10  | sDSb         | 22  | DstBytes     | 34  | DstGap       | 46  | SynAck                    |
| 11  | dDSb         | 23  | Offset       | 35  | Rate         | 47  | AckDat                    |
| 12  | sTtl         | 24  | sMeanPktSz   | 36  | SrcRate      | 48  | Target(anomaly detection) |

**Table 3**

2017 Attack Dataset for Comparison of features across three datasets, features in blue represent common attributes.

| No. | Feature Name                | No. | Feature Name      | No. | Feature Name           | No. | Feature Name              |
|-----|-----------------------------|-----|-------------------|-----|------------------------|-----|---------------------------|
| 1   | Destination Port            | 21  | Fwd IAT Total     | 41  | Packet Length Mean     | 61  | Bwd Avg Packets/Bulk      |
| 2   | Flow Duration               | 22  | Fwd IAT Mean      | 42  | Packet Length Std      | 62  | Bwd Avg Bulk Rate         |
| 3   | Total Fwd Packets           | 23  | Fwd IAT Std       | 43  | Packet Length Variance | 63  | Subflow Fwd Packets       |
| 4   | Total Backward Packets      | 24  | Fwd IAT Max       | 44  | FIN Flag Count         | 64  | Subflow Fwd Bytes         |
| 5   | Total Length of Fwd Packets | 25  | Fwd IAT Min       | 45  | SYN Flag Count         | 65  | Subflow Bwd Packets       |
| 6   | Total Length of Bwd Packets | 26  | Bwd IAT Total     | 46  | RST Flag Count         | 66  | Subflow Bwd Bytes         |
| 7   | Fwd Packet Length Max       | 27  | Bwd IAT Means     | 47  | PSH Flag Count         | 67  | Init_Win_bytes_forward    |
| 8   | Fwd Packet Length Min       | 28  | Bwd IAT Stds      | 48  | ACK Flag Count         | 68  | Init_Win_bytes_backward   |
| 9   | Fwd Packet Length Mean      | 29  | Bwd IAT Max       | 49  | URG Flag Count         | 69  | act_data_pkt_fwd          |
| 10  | Fwd Packet Length Std       | 30  | Bwd IAT Min       | 50  | CWE Flag Count         | 70  | min_seg_size_forward      |
| 11  | Bwd Packet Length Max       | 31  | Fwd PSH Flags     | 51  | ECE Flag Count         | 71  | Active Mean               |
| 12  | Bwd Packet Length Min       | 32  | Bwd PSH Flags     | 52  | Down/Up Ratio          | 72  | Active Std                |
| 13  | Bwd Packet Length Mean      | 33  | Fwd URG Flags     | 53  | Average Packet Size    | 73  | Active Max                |
| 14  | Bwd Packet Length Std       | 34  | Bwd URG Flags     | 54  | Avg Fwd Segment Size   | 74  | Active Min                |
| 15  | Flow Bytes/s                | 35  | Fwd Header Length | 55  | Avg Bwd Segment Size   | 75  | Idle Mean                 |
| 16  | Flow Packets/s              | 36  | Bwd Header Length | 56  | Fwd Header Length      | 76  | Idle Std                  |
| 17  | Flow IAT Mean               | 37  | Fwd Packets/s     | 57  | Fwd Avg Bytes/Bulk     | 77  | Idle Max                  |
| 18  | Flow IAT Std                | 38  | Bwd Packets/s     | 58  | Fwd Avg Packets/Bulk   | 78  | Idle Min                  |
| 19  | Flow IAT Max                | 39  | Min Packet Length | 59  | Fwd Avg Bulk Rate      | 79  | Target(anomaly detection) |
| 20  | Flow IAT Min                | 40  | Max Packet Length | 60  | Bwd Avg Bytes/Bulk     |     |                           |

is supported by metadata like attacker IP addresses (e.g., 10.41.150.68) and timestamps. For this study, a relevant subset of the data focusing on DDoS attacks was utilized, comprising **225,745 samples**. Each sample is described by **78 features** plus a label. The complete list of features used is detailed in [Table 3](#).

#### 4.1.3. 5G Dataset

5G Dataset was generated in a controlled laboratory environment simulating a 5G network. In this environment, DDoS attack traffic was generated against nodes in the network. Each record contains a set of numerical features summarizing the behavior of the flow ( number of packets, total bytes, flow duration, and statistical characteristics of packet sizes). Finally, each flow was labeled as either benign or DDoS attack according to the corresponding scenario and released as pre-processed data files. This dataset provides **686,021 samples**. The original data contains 60 columns; after removing two redundant label-related columns ('malicious' and 'attack number'), **56 features** remain for analysis. These features are specific to 5G core network operations and User Equipment behavior, as shown in [Table 4](#).

#### 4.2. Implementation and hyperparameters

##### 4.2.1. Autoencoder training

Synthetic sample creation and deep feature learning were performed using a fully connected autoencoder with symmetric decoder and encoder networks. The encoder projects the input from 120 dimensions down to 80 dimensions and then to a 40-dimensional latent space. The decoder is its inverse, de-scaling the latent representation out by 80 dimensions and then up to the original 120 dimensions. Rectified Linear

Unit (ReLU) activations are employed in all layers. The input dimension of 120 has both numeric and one-hot encoded categorical attributes. The Adam optimizer was utilized to train the model with an initial learning rate of 0.001 and batch size of 256. Training would stabilize in 50–100 epochs, and early stopping was used to prevent overfitting. Learned latent representations were used for synthesizing synthetic samples and feature transformation in downstream classification tasks.

##### 4.2.2. XGBoost Configuration

The synthetic and original samples were then passed through a supervised learning XGBoost classifier after feature extraction. Hyperparameter tuning was done by grid search over specified ranges, listed in [Table 5](#). The learning rate ( $\eta$ ) was optimized between 0.01 and 0.30 in increments of 0.05 to regulate the rate of convergence. Tree depth was capped at 6 to achieve a balance between model complexity and overfitting. L1 (reg\_alpha) and L2 (reg\_lambda) regularization terms were set over wide ranges to encourage generalization. Moreover, subsampling at the row (Subsample) and column (Column Subsample) levels were included to introduce randomness and minimize variance.

#### 5. Results and discussion

[Table 6](#) depicts the detailed performance metrics of the proposed Hybrid Autoencoder-XGBoost IDS under various training and testing scenarios. Complementing this, [Fig. 2](#) presents a holistic, multi-axis comparison of the key performance and efficiency trade-offs. The metrics of accuracy, F1-score, and Area Under the ROC Curve (AUC) collectively demonstrate the model's effective capability to detect intrusions in varied network traffic.

**Table 4**

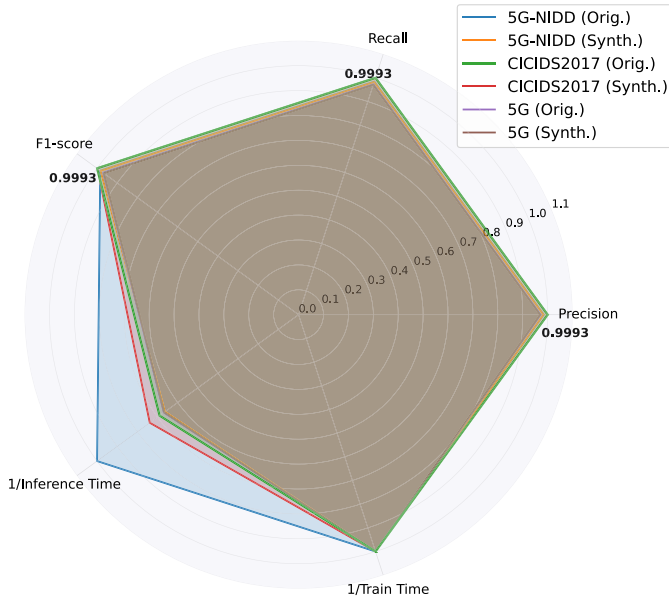
5G Attack Dataset for Comparison of features across three datasets, features in blue represent common attributes.

| No. | Feature Name            | No. | Feature Name            | No. | Feature Name                | No. | Feature Name      |
|-----|-------------------------|-----|-------------------------|-----|-----------------------------|-----|-------------------|
| 1   | time                    | 15  | bearer 1 ip             | 29  | ue aggregate max bitrate dl | 43  | dl retx           |
| 2   | imei                    | 16  | bearer 1 pdu session id | 30  | ue aggregate max bitrate ul | 44  | dl tx             |
| 3   | 5g tmsi                 | 17  | bearer 1 qos flow id    | 31  | ul bitrate                  | 45  | cqi               |
| 4   | amf ue id               | 18  | bearer 1 sst            | 32  | bearer 1 ipv6               | 46  | epre              |
| 5   | bearer 0 apn            | 19  | bearer 1 ul total bytes | 33  | cell                        | 47  | initial ta        |
| 6   | bearer 0 dl total bytes | 20  | dl bitrate              | 34  | ul retx                     | 48  | p ue              |
| 7   | bearer 0 ip             | 21  | ran id                  | 35  | ul err                      | 49  | pusch snr         |
| 8   | bearer 0 ipv6           | 22  | ran plmn                | 36  | ul mcs Min                  | 50  | ri                |
| 9   | bearer 0 pdu session id | 23  | ran ue id               | 37  | ul n layer                  | 51  | turbo decoder avg |
| 10  | bearer 0 qos flow id    | 24  | registere               | 38  | ul path loss                | 52  | turbo decoder max |
| 11  | bearer 0 sst            | 25  | rnti                    | 39  | ul phr                      | 53  | turbo decoder min |
| 12  | bearer 0 ul total bytes | 26  | t3512                   | 40  | ul rank                     | 54  | ul tx             |
| 13  | bearer 1 apn            | 27  | tac                     | 41  | dl err                      | 55  | cell id           |
| 14  | bearer 1 dl total bytes | 28  | tac plmn                | 42  | dl mcs                      | 56  | DDoS attack       |

**Table 5**

XGBoost hyperparameter ranges and increment steps.

| Hyperparameter           | Range     | Step | Role               |
|--------------------------|-----------|------|--------------------|
| Learning Rate ( $\eta$ ) | 0.01–0.30 | 0.05 | Regulates update   |
| Max Depth                | 6         | 1    | Model complexity   |
| L1 Reg. (reg_alpha)      | 0–10      | 1    | Regularization     |
| L2 Reg. (reg_lambda)     | 0–100     | 1    | Regularization     |
| Subsample                | 0.5–1.0   | 0.1  | Data randomness    |
| Column Subsample         | 0.6–1.0   | 0.1  | Feature randomness |

**Fig. 2.** Radar chart comparing the Hybrid Autoencoder-XGBoost IDS across 5G-NIDD, CICIDS2017, and 5G datasets with original (Orig.) and synthetic (Synth.).

### 5.1. Overall performance

The quantitative findings presented in Table 6 indicate the effectiveness of the proposed model. When trained and assessed on the original CIC-IDS2017 samples (row 1), the model achieves high performance metrics, with an accuracy and F1-score of 0.9993 and an AUC of 0.9999. This indicates robust precision and recall. Testing on synthetically created CIC-IDS2017 data (row 2) produces comparably strong results (0.9976 F1-score, 0.9998 AUC), validating the model's adaptability to distribution-matching synthetic data. In the case of the 5G-NIDD dataset (row 3), the model continues to perform well with a 0.9836 F1-score and 0.9991 AUC. The synthetic 5G-NIDD test (row 4) shows only a minor decline (0.9819 F1-score). To conclude, for the 5G dataset (rows 5

and 6), the model achieves F1-scores of 0.9695 for the original data and 0.9658 for the synthetic data. The anticipated minor decrease in performance on the 5G-specific datasets can be explained by the differences in domains between conventional IP traffic and 5G control-plane traffic. Nonetheless, the overall outcomes still demonstrate a high level of performance. Finally, the last row of Table 6 presents a leave-one-dataset-out evaluation, where the 5G dataset is completely excluded from training and used only for testing. Although performance in this setting is slightly lower than in mixed-training evaluations, this behavior is expected due to domain shifts arising from differences in data collection environments, traffic patterns, and network conditions, and provides a more realistic assessment of cross-dataset generalization.

The radar chart displayed in Fig. 2 visually supports these results. The findings clearly demonstrate that all experimental scenarios yield results that are tightly clustered around the optimal value of 1.0 on the Precision, Recall, and F1-score axes, confirming a high level of classification accuracy overall. The graphs for both the original (Orig.) and synthetic (Synth.) samples across each dataset exhibit a close grouping, indicating that the synthetic data effectively maintains performance integrity. The chart also effectively illustrates the primary trade-off: computational cost. The 5G-NIDD (Orig.) line shows a noticeably lower value on the "1/Inference Time" axis, indicating a longer inference time (higher computational cost) for that scenario compared to the others.

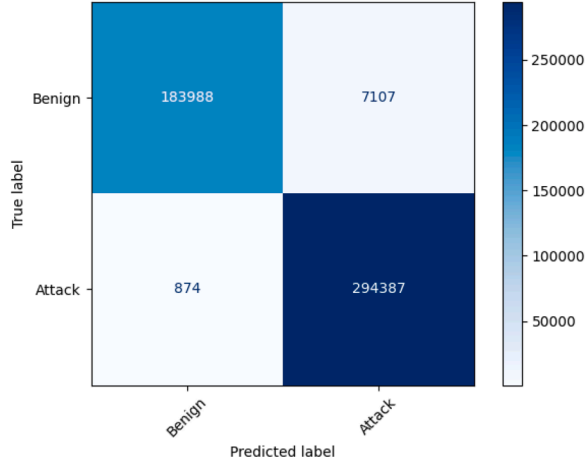
Fig. 3 illustrates the confusion matrices used to evaluate the zero-day attack scenario on the 5G-NIDD dataset. In this experiment, the TCP Connection Scan attack is completely excluded from the training phase and introduced only during testing to simulate an unseen attack type. Sub-Figs. 3 (a) and 3 (c) show the baseline performance without zero-day traffic for the synthetic and standard settings, respectively. Sub-Figs. 3 (b) and 3 (d) present the corresponding results when the zero-day attack is included in the test set. The sub-diagonal elements of the confusion matrices represent misclassification errors. Comparing the baseline and zero-day cases reveals an increase in prediction errors when the unseen attack is present, indicating the model's sensitivity to previously unseen attack behavior under an offline evaluation setting.

### 5.2. Impact of synthetic augmentation

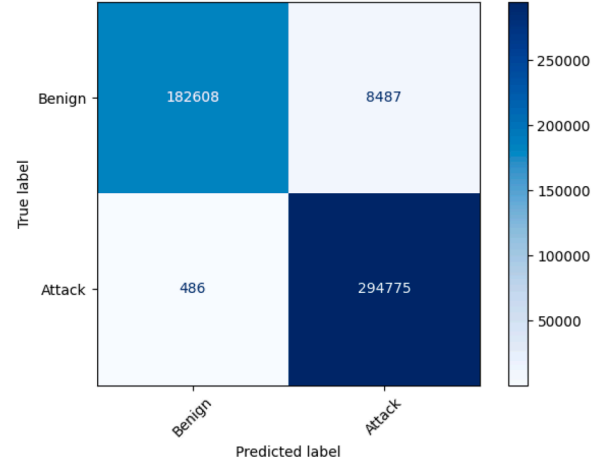
The synthetic flows are generated from the latent representations learned by the stacked autoencoder, which compresses each packet into a dense, noise-robust vector space. Augmenting the training data in that latent space offers two complementary benefits: it counters class imbalance and injects controlled variability that regularises the downstream XGBoost learner, curbing over-fitting. The high quality of this synthetic data is evident when comparing the model's performance on original versus synthetic test sets, as detailed in Table 7. For instance, when testing on CIC-IDS2017 data, the average F1-score for original samples is 0.9993 (row 1), while the score for synthetic samples is a very comparable 0.9972 (row 2).

**Table 6**  
Performance Under Various Training and Testing Conditions.

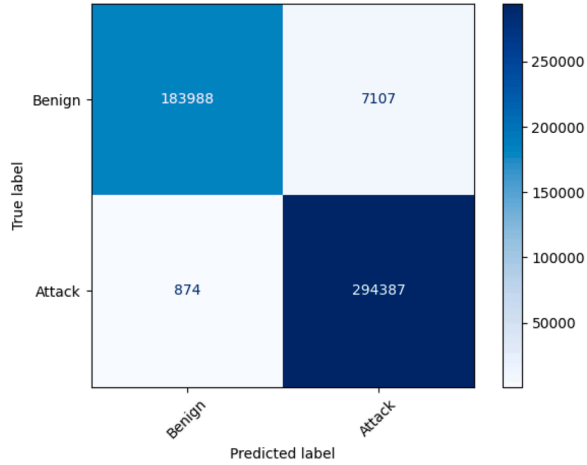
| Train Data (Original + Test Data Synthetic) | Test Data                       | Accuracy | precision | recall | F1 Score |
|---------------------------------------------|---------------------------------|----------|-----------|--------|----------|
| CIC-IDS2017 + 5G-NIDD + 5G                  | CIC-IDS2017 (Original samples)  | 0.9993   | 0.9992    | 0.9992 | 0.9993   |
| CIC-IDS2017 + 5G-NIDD + 5G                  | CIC-IDS2017 (Synthetic samples) | 0.9976   | 0.9976    | 0.9976 | 0.9976   |
| CIC-IDS2017 + 5G-NIDD + 5G                  | 5G-NIDD (Original samples)      | 0.9836   | 0.9838    | 0.9835 | 0.9836   |
| CIC-IDS2017 + 5G-NIDD + 5G                  | 5G-NIDD (Synthetic samples)     | 0.9820   | 0.9823    | 0.9819 | 0.9819   |
| CIC-IDS2017 + 5G-NIDD + 5G                  | 5G (Original samples)           | 0.9724   | 0.9713    | 0.9723 | 0.9695   |
| CIC-IDS2017 + 5G-NIDD + 5G                  | 5G (Synthetic samples)          | 0.9697   | 0.9690    | 0.9697 | 0.9658   |
| CIC-IDS2017 + 5G-NIDD                       | 5G                              | 0.9386   | 0.8810    | 0.9386 | 0.9089   |



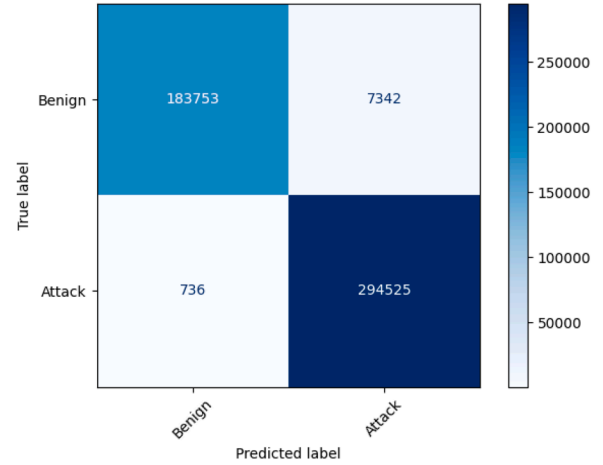
(a) 5G-NIDD-synth.



(b) 5G NIDD synth ZeroDay (without TCPConnectScan in Train).



(c) 5G NIDD standard.



(d) 5G NIDD standard ZeroDay (without TCPConnectScan in Train).

**Fig. 3.** Confusion matrices for offline zero-day attack evaluation on the 5G-NIDD dataset, comparing standard and synthetic settings with and without the TCP Connection Scan attack excluded from training.

**Table 7**  
F1 score per fold in 5-fold cross-validation.

| Train Data              | Test Data           | Fold1  | Fold2  | Fold3  | Fold4  | Fold5  | Average | STD     | Var     |
|-------------------------|---------------------|--------|--------|--------|--------|--------|---------|---------|---------|
| CICIDS2017+5G-NIDD + 5G | CICIDS2017 (Orig.)  | 0.9992 | 0.9996 | 0.9993 | 0.9994 | 0.9993 | 0.9993  | 1.35e-4 | 1.84e-8 |
| CICIDS2017+5G-NIDD + 5G | CICIDS2017 (Synth.) | 0.9969 | 0.9966 | 0.9977 | 0.9982 | 0.9965 | 0.9972  | 6.61e-4 | 4.37e-7 |
| CICIDS2017+5G-NIDD + 5G | 5G-NIDD (Orig.)     | 0.9832 | 0.9839 | 0.9835 | 0.9835 | 0.9841 | 0.9837  | 3.19e-4 | 1.02e-7 |
| CICIDS2017+5G-NIDD + 5G | 5G-NIDD (Synth.)    | 0.9822 | 0.9817 | 0.9817 | 0.9824 | 0.9843 | 0.9821  | 9.60e-4 | 9.22e-7 |
| CICIDS2017+5G-NIDD + 5G | 5G (Orig.)          | 0.9695 | 0.9695 | 0.9702 | 0.9701 | 0.9692 | 0.9697  | 3.84e-4 | 1.47e-7 |
| CICIDS2017+5G-NIDD + 5G | 5G (Synth.)         | 0.9676 | 0.9651 | 0.9645 | 0.9635 | 0.9668 | 0.9655  | 1.50e-3 | 2.25e-6 |

*Note:* Train data combine original and synthetic samples from the listed datasets; test data use the subset indicated in parentheses. **STD**= standard deviation, **Var**= variance.

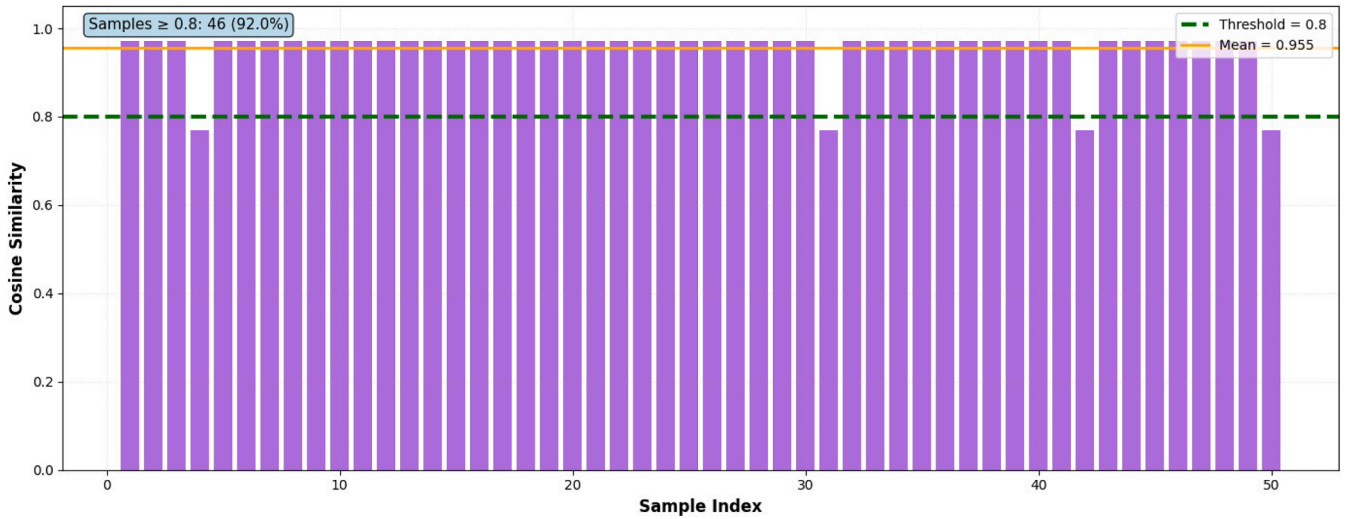


Fig. 4. Cosine similarity analysis between synthetic and original attack samples from the CICIDS2017 dataset, showing the 0.8 similarity threshold and mean similarity.

A similar pattern is observed for the 5G-NIDD dataset, with an average F1-score of 0.9837 on original samples (row 3) and 0.9821 on synthetic samples (row 4). This minimal performance difference confirms that the synthetic samples are realistic and effective, helping the model achieve effective class separability (as also indicated by the high AUC values in your main performance table) without degrading performance. In addition, to further examine the behavioral consistency between generated synthetic samples and original attack traffic, a subset of 50 DDoS samples was randomly selected from the CICIDS2017 dataset, and 50 corresponding synthetic samples were subsequently generated. Cosine similarity analysis was then used to assess representational similarity between original and synthetic samples. As illustrated in Fig. 4, a similarity threshold of 80 percent was used to quantify behavioral alignment. The X-axis represents cosine similarity values, while blue bars indicate similarities above the 80 percent threshold and red bars denote samples below this threshold. The results show that the majority of synthetic samples preserve feature-space characteristics consistent with the underlying attack behavior.

These observations confirm that coupling deep representation learning with gradient-boosted classification, augmented by synthetic data, yields a practical, scalable solution for securing heterogeneous 5G environments. The consistently high AUCs indicate that the learnt embeddings capture protocol-agnostic properties of malicious behavior, enabling seamless transfer from wired IDS benchmarks to emerging 5G control-plane applications ranging from industrial IoT to consumer mobile services.

#### 5.2.1. Model stability and efficiency

To further demonstrate the model's stability, a five-fold cross-validation scheme was conducted. The results, fully elaborated in Table 7, show the F1-score calculated for each fold on all combinations of test sets. The column labeled 'Average' exhibits high consistency and has very low inter-fold variance, thereby supporting the reliability of the model. An example is that the average F1-score of the original CICIDS2017 test data is 0.9993, and fold scores are almost within the range of 0.9992 to 0.9996. This stability is always noted in all assessed test cases.

Table 8 presents the computational costs associated with the 5-fold cross-validation, measured in wall-clock seconds. The 'Training only' row shows the time required for model training in each fold, averaging approximately 57.5 seconds. The other rows detail the inference (testing) time and the time required to generate SHAP explanations for different test sets. The inference times are consistently low, often un-

der one second for most test sets, which highlights the efficiency of the lightweight autoencoder-XGBoost pipeline and its suitability for real-time anomaly detection.

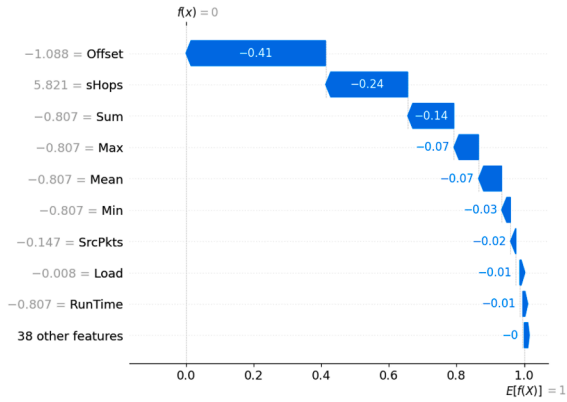
#### 5.3. Model explainability

Model explainability refers to the capacity to comprehend and interpret how a machine learning model generates predictions. It is especially important in areas where the implications of model mistakes or prejudice have tremendous implications, i.e., medicine, finance, and IDS. Explainability also prevents models from turning into black boxes; hence, high classification accuracy is complemented by transparency and interpretability. In this study, SHAP is used to explain global and local interpretability of the learned model. SHAP, introduced by Lundberg et al. [37], is a model-agnostic explanation technique that produces generic explanations irrespective of the classifier used. SHAP values indicate the value of a single feature's contribution to the output of the model and thus allow improved understanding of how individual attributes affect predictions [38–40].

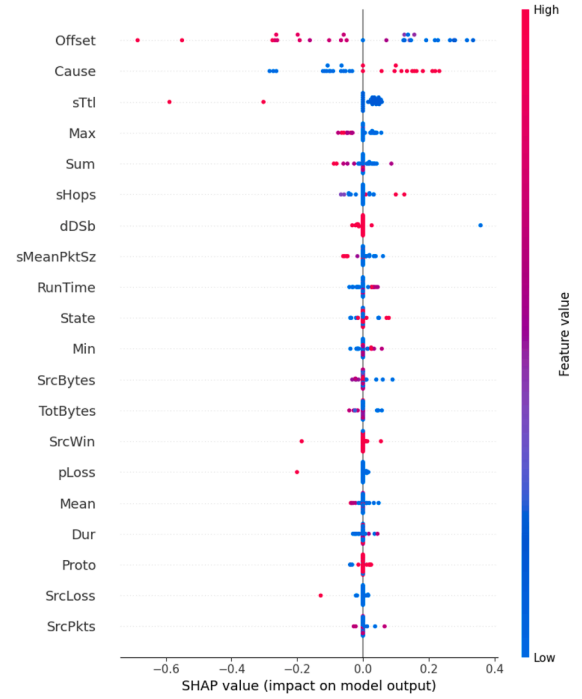
As Fig. 5 (a), (c), and (e) show, local explanations unveil how single features influence the classifier's prediction for sample flows. For the 5G-NIDD example (Fig. 5 (a)), features like *Offset* and *Sum* have a strong negative impact, pushing the prediction towards the benign class. By contrast, the CIC-IDS2017 instance (Fig. 5 (c)) displays a model of consistently positive contributions from features like *min\_seg\_size\_forward* and *Min Packet Length*, pointing sharply toward an attack classification. Finally, the 5G dataset local explanation in Fig. 5 (e) shows that 5G-specific features like *turbo\_decoder\_avg*, *bearer\_0\_ul\_total\_bytes*, and *amf\_ue\_id* are the primary contributors pushing the model to flag a flow as malicious.

The global summary plots in Fig. 5 (b), (d), and (f) cumulate these feature impacts. In the 5G-NIDD case (Fig. 5 (b)), top predictors are *Offset*, *Cause*, and *sTtl*. High values of *sTtl* (red dots) tend predictions towards benign (negative SHAP value), as expected from legitimate control packets. For the CIC-IDS2017 dataset (Fig. 5 (d)), *ACK Flag Count* and *Min Packet Length* are dominant, with high values strongly correlating with attack streams. The global summary for the 5G dataset (Fig. 5 (f)) reveals a different set of key features, including *bearer\_0\_ip*, *bearer\_0\_apn*, and *d1\_retx* (downlink retransmissions). High values for these bearer-specific attributes and data plane metrics are strongly correlated with attack predictions, demonstrating the model's ability to leverage granular 5G core network features to identify threats.

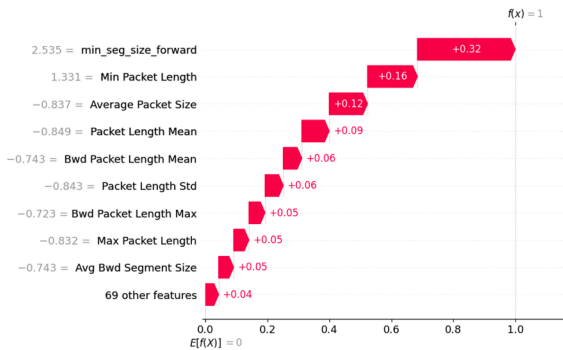




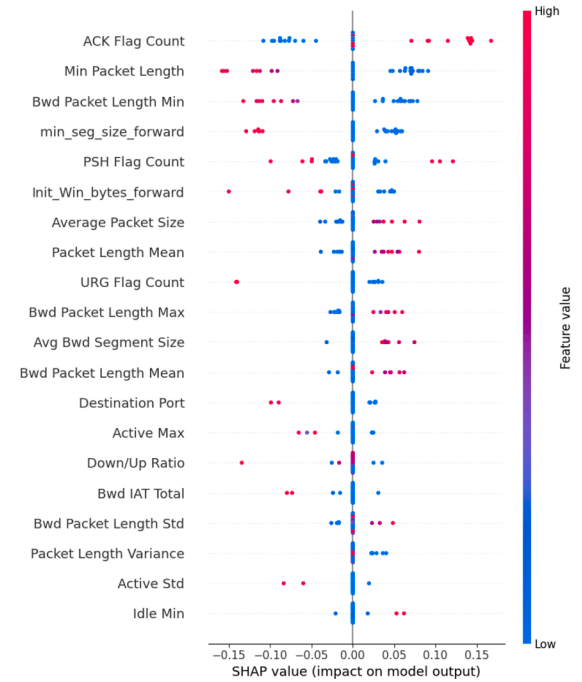
(a) Contribution of Features for 5G-NIDD



(b) SHAP summary plot for the first 30 samples from the test set of the 5G-NIDD dataset. The selected samples include both benign and attack instances.

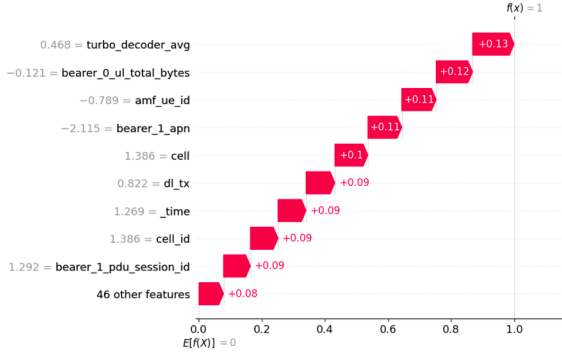


(c) Contribution of Features for CIC-IDS2017



(d) SHAP summary plot for the first 30 samples from the test set of the CIC-IDS2017 dataset. The selected samples include both benign and attack instances.

Fig. 5. SHAP results for Hybrid Autoencoder-XGBoost IDS under various scenarios.



(e) Contribution of Features for 5G



(f) SHAP summary plot for the first 30 samples from the test set of the 5G dataset. The selected samples include both benign and attack instances.

Fig. 5. SHAP results for Hybrid Autoencoder-XGBoost IDS under various scenarios. (Continued).

This detailed SHAP analysis provides crucial insights into the model's decision-making. The model predictions are heavily influenced by feature categories specific to each network type. For traditional IP traffic (CIC-IDS2017), it relies on payload size (Min Packet Length) and TCP flags (ACK Flag Count). For 5G-NIDD traffic, it leverages routing metadata (specifically sTt1). This aligns perfectly with the realities of 5G control-plane traffic, where legitimate packets follow standard paths, leading to predictable sTt1 values. As Fig. 5 (b) shows, high sTt1 (red dots) pushes the prediction toward benign (negative SHAP value). Conversely, attack traffic, which may be spoofed, often presents with non-standard TTLs. Therefore, the model has learned to associate deviations from standard packet sizes and TTLs with a high probability of intrusion. Beyond visual inspection, the SHAP explanations are intended for direct operational use by security analysts. In our envisioned deployment, each alert generated by the AE-XGBoost IDS is accompanied by a local SHAP explanation that highlights the small set of features that most strongly contributed to the decision (e.g., sTt1 and Cause for 5G-NIDD, or ACK Flag Count and Min Packet Length for CIC-IDS2017). This al-

lows analysts to (i) justify immediate mitigation actions, (ii) quickly dismiss false positives when the dominant features match known benign patterns, and (iii) derive new signature-like rules based on frequently recurring high-impact features. Global SHAP summaries, aggregated over longer periods, guide policy refinement by revealing which 5G-specific KPIs such as retransmissions (dl\_retx) or turbo-decoder statistics (turbo\_decoder\_avg) consistently drive attack predictions. In this way, SHAP moves beyond presentation and becomes an operational tool that supports triage, rule engineering, and model governance in B5G/6G security operations centers.

#### 5.4. Comparative analysis of classifier performance

To validate the selection of XGBoost as the classification component, we conducted a comparative analysis against other standard machine learning models: Decision Tree (AE-DT), Random Forest (AE-RF), and K-Nearest Neighbors (AE-KNN). These baselines are our own implementations of Traditional classifiers [29–31], instantiated on top of the

Table 8

Time per fold in five-fold cross-validation.

| Train Data (Original + Synthetic) | Test Data                       | Fold1 [s] | Fold2 [s] | Fold3 [s] | Fold4 [s] | Fold5 [s] |
|-----------------------------------|---------------------------------|-----------|-----------|-----------|-----------|-----------|
| CIC-IDS2017 + 5G-NIDD + 5G        | CIC-IDS2017 (Original samples)  | 0.15      | 0.15      | 0.15      | 0.31      | 0.31      |
| CIC-IDS2017 + 5G-NIDD + 5G        | CIC-IDS2017 (Synthetic samples) | 0.15      | 0.15      | 0.17      | 0.22      | 0.30      |
| CIC-IDS2017 + 5G-NIDD + 5G        | 5G-NIDD (Original samples)      | 0.83      | 0.82      | 0.85      | 1.63      | 1.63      |
| CIC-IDS2017 + 5G-NIDD + 5G        | 5G-NIDD (Synthetic samples)     | 1.37      | 0.83      | 0.85      | 1.59      | 1.59      |
| CIC-IDS2017 + 5G-NIDD + 5G        | 5G (Original samples)           | 0.46      | 0.46      | 0.64      | 0.87      | 0.92      |
| CIC-IDS2017 + 5G-NIDD + 5G        | 5G (Synthetic samples)          | 0.47      | 0.45      | 0.09      | 0.88      | 0.92      |
| CIC-IDS2017 + 5G-NIDD + 5G        | Training only                   | 46.13     | 45.36     | 44.50     | 74.46     | 77.36     |
| CIC-IDS2017 + 5G-NIDD + 5G        | SHAP 5G-NIDD                    | 0.71      | 0.76      | 1.01      | 1.01      | 0.94      |
| CIC-IDS2017 + 5G-NIDD + 5G        | SHAP 5G                         | 0.80      | 0.78      | 1.09      | 1.11      | 1.01      |
| CIC-IDS2017 + 5G-NIDD + 5G        | SHAP 2017                       | 1.43      | 1.81      | 2.06      | 1.94      | 2.03      |

Note: All values are expressed in wall-clock seconds. The “Training only” row displays training time, while the other rows show test time and SHAP analysis time.

**Table 9**  
Accuracy of method and scenario.

| Train Data (Original + Test Data Synthetic) | Test Data                       | AE-XGB | AE-DT  | AE-RF         | AE-KNN        |
|---------------------------------------------|---------------------------------|--------|--------|---------------|---------------|
| CIC-IDS2017 + 5G-NIDD + 5G                  | CIC-IDS2017 (Original samples)  | 0.9993 | 0.9981 | <b>0.9997</b> | 0.9995        |
| CIC-IDS2017 + 5G-NIDD + 5G                  | CIC-IDS2017 (Synthetic samples) | 0.9976 | 0.9989 | <b>0.9995</b> | 0.9990        |
| CIC-IDS2017 + 5G-NIDD + 5G                  | 5G-NIDD (Original samples)      | 0.9836 | 0.9832 | 0.9838        | <b>0.9852</b> |
| CIC-IDS2017 + 5G-NIDD + 5G                  | 5G-NIDD (Synthetic samples)     | 0.9820 | 0.9838 | 0.9842        | <b>0.9845</b> |
| CIC-IDS2017 + 5G-NIDD + 5G                  | 5G (Original samples)           | 0.9723 | 0.9663 | <b>0.9776</b> | 0.9759        |
| CIC-IDS2017 + 5G-NIDD + 5G                  | 5G (Synthetic samples)          | 0.9697 | 0.9677 | 0.9779        | <b>0.9783</b> |

**Table 10**  
Test Time of method and scenario.

| Train Data (Original + Test Data Synthetic) | Test Data                       | AE-XGB | AE-DT         | AE-RF  | AE-KNN |
|---------------------------------------------|---------------------------------|--------|---------------|--------|--------|
| CIC-IDS2017 + 5G-NIDD + 5G                  | CIC-IDS2017 (Original samples)  | 0.3087 | <b>0.0292</b> | 0.9707 | 864    |
| CIC-IDS2017 + 5G-NIDD + 5G                  | CIC-IDS2017 (Synthetic samples) | 0.3305 | <b>0.0157</b> | 0.9835 | 2319   |
| CIC-IDS2017 + 5G-NIDD + 5G                  | 5G-NIDD (Original samples)      | 2.4108 | <b>0.2747</b> | 6.8178 | 4660   |
| CIC-IDS2017 + 5G-NIDD + 5G                  | 5G-NIDD (Synthetic samples)     | 1.5994 | <b>0.0873</b> | 6.1775 | 12564  |
| CIC-IDS2017 + 5G-NIDD + 5G                  | 5G (Original samples)           | 0.9111 | <b>0.0698</b> | 4.0089 | 2755   |
| CIC-IDS2017 + 5G-NIDD + 5G                  | 5G (Synthetic samples)          | 0.9078 | <b>0.0454</b> | 4.7290 | 6880   |

same autoencoder backbone as the proposed AE-XGB model. All models were trained on the same rich features extracted by the autoencoder (AE). The comparison focused on the critical trade-off between detection accuracy (Table 9) and inference speed (Table 10). As shown in Table 9, all classifiers achieve high detection accuracy. The AE-RF and AE-KNN models frequently post the highest scores; for instance, AE-RF achieves 0.9997 on the original CIC-IDS2017 samples, while AE-KNN reaches 0.9852 on the original 5G-NIDD set. However, the proposed AE-XGB model remains competitive, with accuracy scores that are only fractionally lower than the top performers (e.g., 0.9836 vs. 0.9852 on the 5G-NIDD set).

This minor difference in accuracy is contrasted by a significant divergence in computational cost, as detailed in Table 10. The AE-KNN model, despite its high accuracy, is computationally prohibitive for a real-world IDS, with inference times scaling to thousands of seconds (e.g., 12,564 seconds for the synthetic 5G-NIDD test). AE-RF shows a significantly slower performance as compared to AE-XGB. On the other hand, AE-DT is the quickest to infer; nevertheless, as Table 9 elaborates, it consistently yields the poorest performance in terms of accuracy compared to the other models studied. Such analysis, therefore, highlights a typical trade-off between accuracy and efficiency. In line with this, the AE-XGB model is placed as the best option. Its accuracy is highly reliable and comparable to that of the best-performing models (AE-RF, AE-KNN), and it offers the second-fastest inference speeds compared to all other models, except those that are less accurate, such as AE-DT. This trade-off between high accuracy and high computational efficiency makes AE-XGBoost the most realistic and viable classifier in this hybrid IDS architecture. It is also instructive to relate these results to representative intrusion detection systems reported in the literature. For example, on the CIC-IDS2017 dataset, our AE-XGB model achieves 0.9993 accuracy and F1-score with an AUC of 0.9999, whereas the explainable ensemble of RF, DT, and SVM analyzed in [28] reports around 96.25% accuracy on the same dataset. While the exact experimental setups differ and direct numerical comparisons must therefore be interpreted with caution, this suggests that the proposed framework is competitive with, and often exceeds, the detection performance of existing XAI-based IDS solutions. Furthermore, prior autoencoder-XGBoost IDS models [21,23–25] have largely been evaluated on pre-5G benchmarks such as NSL-KDD or CSE-CIC-IDS2018, and typically do not combine latent-space synthetic augmentation with integrated SHAP explanations and cross-dataset validation on multiple 5G/B5G datasets. In contrast, our AE-XGB pipeline jointly addresses high-dimensional 5G traffic, severe class imbalance, and operational explainability in a single framework.

## 6. Conclusion and future directions

This paper presents an explainable hybrid Intrusion Detection System designed to meet the security needs of 5G networks. By integrating deep autoencoder-based representation learning, synthetic data augmentation, and XGBoost classification into a cohesive pipeline, it effectively addresses critical challenges such as high-dimensional traffic, class imbalance, and the detection of low-profile attacks. The framework is evaluated on three datasets, including CIC-IDS2017, 5G-NIDD, and the 5G DDoS dataset under 5-fold cross-validation and cross-dataset experiments. The main findings of our analysis can be summarized as follows. First, the proposed model achieves consistently high detection performance across all evaluated datasets, with F1-scores ranging from approximately 0.96 to 0.999 and AUC values above 0.97, indicating strong generalization across heterogeneous traffic sources. Second, the difference between results on original and synthetic test samples is small, which confirms that the latent-space synthetic augmentation produces realistic flows and effectively mitigates class imbalance without sacrificing accuracy. Third, the 5-fold cross-validation results show low variance across folds and inference times that are often below one second for all test sets, while training times remain on the order of tens of seconds, suggesting that the framework is suitable for near real-time deployment in 5G environments. Fourth, SHAP-based explainability reveals that a small subset of 5G- and TCP-level features, dominates the model's decisions and aligns with domain knowledge on benign versus malicious behaviour, providing actionable insights for security analysts. However, as 5G technology continues to advance, the integration of intelligence into its architecture must evolve in tandem with it. This should not entail the wholesale implementation of overly complex models, but rather a careful calibration of capability, efficiency, and trust. Future research should therefore address four intertwined challenges. First, continuous adaptation is required: online and incremental learning routines must allow the autoencoder and XGBoost layers to absorb evolving traffic patterns without service disruption. Second, privacy-preserving scalability is essential; federated or split-learning variants can distribute training across edge sites, retaining raw packet traces locally and satisfying data-sovereignty mandates. Third, advanced data-synthesis techniques, such as conditional or variational autoencoders and diffusion-based generators, can produce protocol-aware attack replicas that bolster minority-class representation without encouraging overfitting. Finally, lightweight, in-line explanation modules are needed to make operators understand in real time why a flow is flagged, thus closing the gap between laboratory accuracy and field trust.

## CRediT authorship contribution statement

**Nasim Nezhadsistani:** Writing – original draft, Validation, Methodology, Formal analysis; **Mohsen Tajgardan:** Software, Formal analysis; **Mahdi Rabbani:** Writing – review & editing, Investigation; **Weijie Niu:** Writing – review & editing; **Burkhard Stiller:** Writing – review & editing, Supervision, Resources.

## Data availability

Data will be made available on request.

## Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Nasim Nezhadsistani reports financial support was provided by Swiss State Secretariat for Education Research and Innovation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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